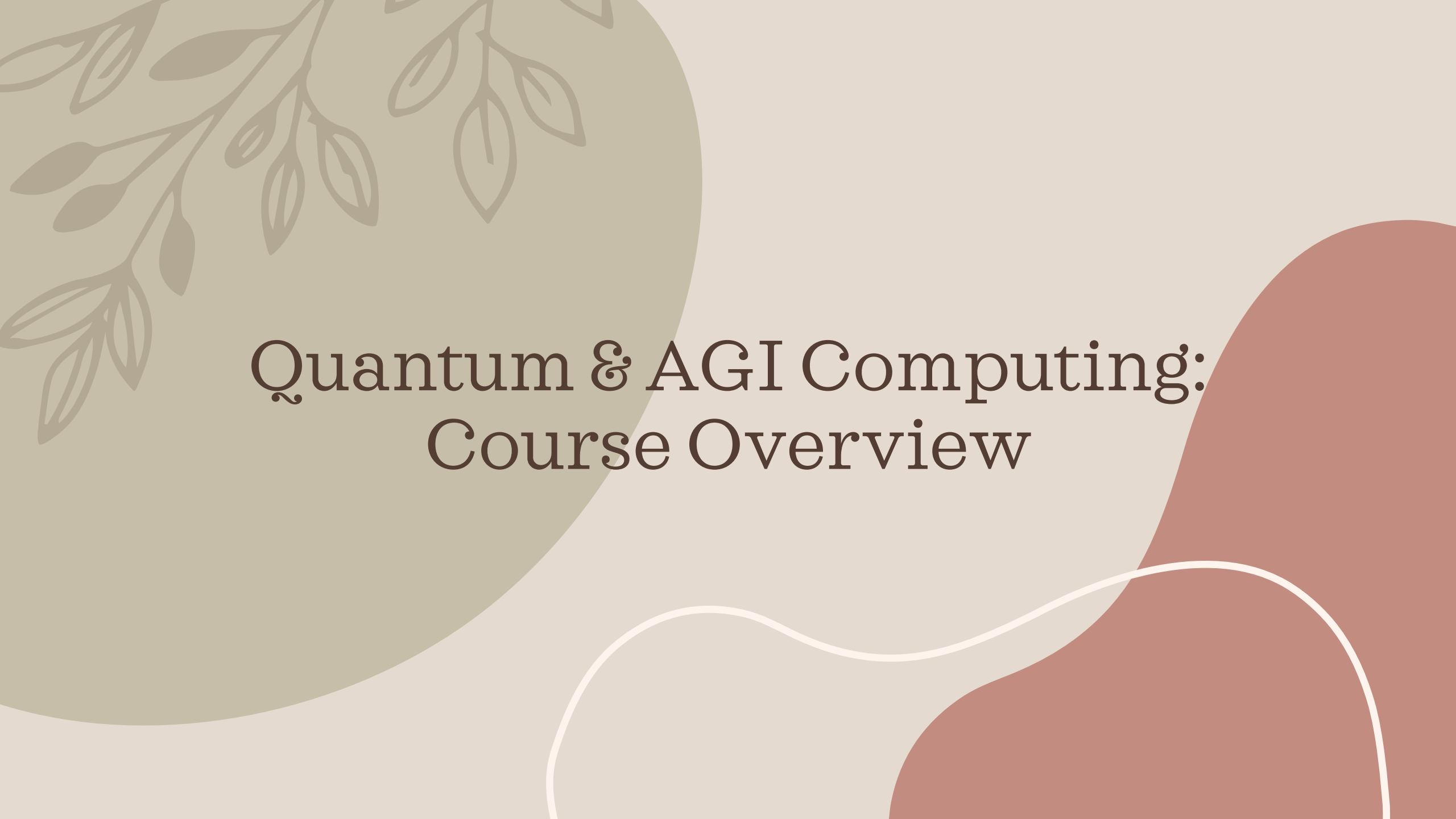




Quantum & AGI Computing: Course Overview

Superposition & Entanglement

- **“Quantum computing” is an emerging approach that represents a significant departure from classical computing approaches.**
- **This new form of computing is rooted in quantum mechanics, which focuses on the functioning of matter at lower scale levels, such as atoms and subatomic particles.**
- **Unlike classical computing, quantum computing utilizes qubits, which differ from the standard bits used in electronic computers.**

Superposition & Entanglement

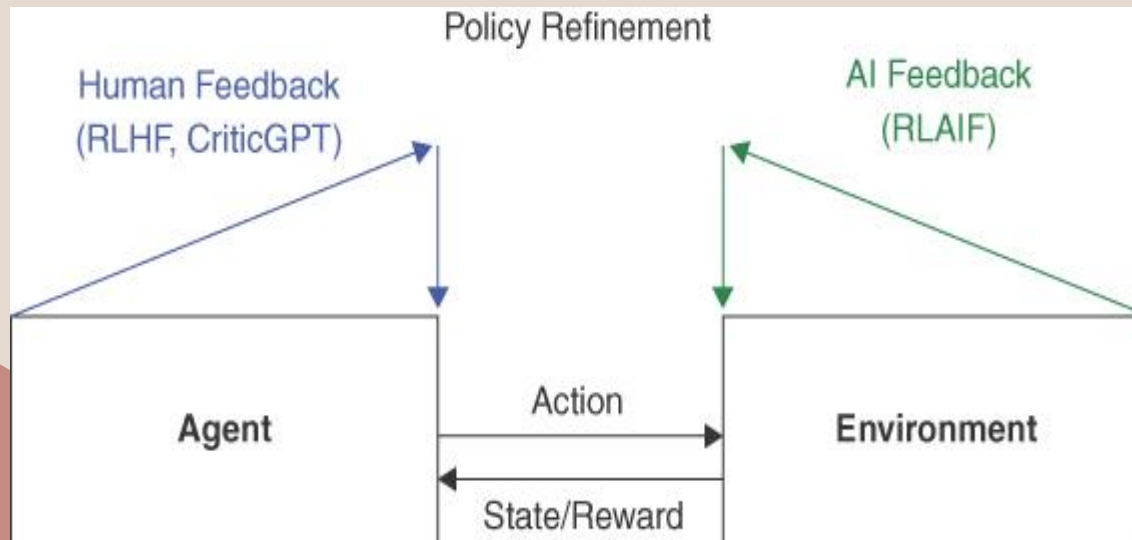
- While regular bits are 0 or 1, qubits can simultaneously exist in both states, a phenomenon known as superposition.
- Superposition enables quantum computers to process vast information rapidly, making them significantly faster than conventional computers.
- Another important aspect of quantum computing is quantum entanglement, where qubits become interconnected uniquely.
- When two quantum objects are entangled, altering one object causes a corresponding change in the other, regardless of the distance between them. This characteristic allows quantum computers to perform numerous calculations concurrently, a feat unattainable by regular computers.

Artificial General Intelligence (AGI)

- **Quantum computing is one of the three main pillars of the Artificial General Intelligence (AGI).**
- **AGI is the next wave of AI where intelligence is applied autonomously and flexibly as if the AI agent is a human being,**
- **In AGI, agents are expected to act, behave and independently take decisions, without the interference or guidance from human beings**
- **The other two pillars are Agentic AI and blockchain.**

AI Agents

- “AI Agents” is a new-old concept that refers to intelligent agents that interact with an environment or another intelligence sources to make decisions
- A good example are the Reinforcement Learning agents which interact with environments to make intelligent decisions and solve problems
- Recently, the term “AI Agents” got a new dimension, with ad advent of “Generative AI Agents”, where the intelligent agents interact with LLM generative systems to solve problems and provide answers



Agentic AI

- The success of AI Agents in Generative AI Chatbots led to the emergence of a new field: Agentic AI
- In agentic AI, Autonomous AI agents are designed and implemented using emerging platforms such as Langchain, AutoGen, Crew.
- *An AI agent is a computational system that perceives its environment, processes internal state, and takes actions to achieve defined goals.*
- *These systems exhibit autonomy, adaptability, and reactivity—key attributes that differentiate them from traditional software programs.*
- *An agent operates not merely by reacting to inputs, but by maintaining context, managing goals, and adjusting strategies based on feedback.*
- *This dynamic behavior draws from the paradigm of **situated AI**, where intelligence emerges from continuous interaction with the environment. “*

Blockchain Technologies

- Since their emergence in 2008, Blockchain system technologies became the Defacto approach for building secure, decentralized, reliable and resilient systems
- At some stage they became known as Web-3 or Value-Web
- Their role is providing trust in the anticipated Artificial General Intelligence

Reference One

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